GRE Quantitative Reasoning

모의고사 Set 12 — 문제지

최신 문제(뒷번호) · S2 고난도

구성
Section 1: 12문항 / 21분
Section 2: 15문항 / 26분
합계 27문항 / 47분 · 계산기 사용 가능

대소비교(Quantitative Comparison) 선택지 — 모든 QC 문제 공통
(A) Quantity A is greater. (B) Quantity B is greater.
(C) The two quantities are equal. (D) The relationship cannot be determined from the information given.

안내
- Indicate all (복수선택) 문제는 해당하는 것을 모두 고르세요. 모두 맞아야 정답입니다.
- 주관식 분수 답은 약분하지 않아도 됩니다.
- 문제 이미지 속 번호는 문제집 원번호입니다. 문항 번호는 보라색 'Question N'을 기준으로 합니다.
- 정답과 해설은 별도 파일(Set 12 해설지)에 있습니다.

답안 기록표

S1	1	2	3	4	5	6	7	8	9	10	11	12
답												

S2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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Section 1 12 Questions · 21 Minutes

Question 1 대소비교

973. The length and width of rectangle R are x and y, respectively, and the diagonal of R has length 12.
Quantity A: $(x+y)^2$
Quantity B: 144

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

894. x is a positive integer.
Quantity A: $(-1)^x$
Quantity B: $(-1)^{-x}$

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

677. Quantity A: The remainder when 7^{13} is divided by 10
Quantity B: 7

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

1073. N is a prime integer greater than 5.
Quantity A: The units digit of N^4
Quantity B: 3

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 5 객관식

509. In a certain club, 40 percent of the members are less than 25 years old and 66 percent of the members are less than 35 years old. Approximately what fraction of the members of the club are at least 25 years old but less than 35 years old?
A1/2
B1/3
C1/4
D1/5
E1/6

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Questions 6 to 8 are based on the following data.

SURVEY RESULTS

Characteristic	Percent
Work experience	72%
Proficiency in English	68%
Ability to follow directions	65%
Specific occupational skill	60%
Computer expertise	58%
Appropriate attire and behavior	55%

A survey of 550 male managers and 650 female managers was conducted. All 1,200 managers identified whether, for each of six characteristics, the characteristic is important to consider when hiring a new employee. For each of the six characteristics, the percent of managers surveyed who identified that characteristic as important to consider is given in the following table.

Question 6 복수선택

667. Which of the following statements about the managers surveyed must be true?
ALess than 55 percent were male managers.
BOf the male managers, more identified work experience as an important characteristic to consider than identified proficiency in English.
CLess than 60 percent of the male managers identified specific occupational skill as important to consider.

Indicate all that apply. [A] [B] [C]

Question 7 객관식

668. The number of managers surveyed who identified work experience as an important characteristic to consider was approximately what percent greater than the number who identified appropriate attire and behavior as an important characteristic to consider?
A15%
B20%
C25%
D30%
E35%

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 8 객관식

669. If 48 percent of the managers surveyed identified both ability to follow directions and computer expertise as an important characteristics to consider, what percent of the managers surveyed identified neither of these characteristics as important to consider?
A15%
B18%
C23%
D25%
E28%

A B C D E

Question 9 객관식

1161. 2h+3k=5, 2j-k=2, what is the average of h, k, and j?
A.7/2
B.7/3
C.7/4
D.7/6
E.?

A B C D E

Question 10 주관식

1320. Triangles ABC has sides of lengths 6,7, and j, where j is an integer. Triangle DEF has sides of lengths 2,13, and j. What is the value of j?

Question 11 복수선택

987. If a > 1, which of the following could be the value of $(1+(1/a))^{-1}$? Indicate all such values.
A0.35
B0.42
C0.57
D0.75
E0.98
F1.06

Indicate all that apply. A B C D E F

Question 12 객관식

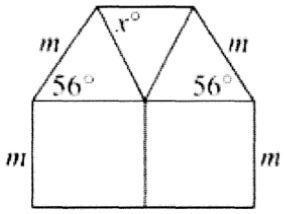
983. A total of $96,000 was invested for one month in a new money market account that paid simple annual interest at the rate of r percent. If the investment earned $960 in interest for the month, what is the value of r?
A5
B7
C12
D13
E14

A B C D E

Section 2 15 Questions · 26 Minutes

Question 1 대소비교

824.



The interior of the figure shown is divided into two squares and three triangles.
Quantity A: The measure of angle x
Quantity B: 56

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 2 대소비교

904. In xy-plane, the graph of y=ax+b, where ab>0, has the y-intercept s and the x-intercept t.
Quantity A: s+t
Quantity B: b

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 3 대소비교

495. x and y are integers such that 0 < y < x.
H=100x+10x+4
G=100y+10y+2
M=(H-G)(H+G)
Quantity A: The units digit of M
Quantity B: 2

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 4 대소비교

847. x<y, The median of x, y and 120 is 100
Quantity A: x
Quantity B: 90

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

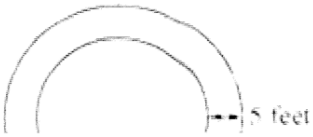
Question 5 대소비교

887. A certain list consists of 25 different positive integers, where each of the integers is a multiple of the positive integer m, The greatest integer in the list is 250.
Quantity A: m
Quantity B: 5

(A) Quantity A is greater.
(B) Quantity B is greater.
(C) The two quantities are equal.
(D) The relationship cannot be determined from the information given.

Question 6 객관식

1175.



The figure represents a 5-foot-wide circular walkway that is bounded by two concentric circles. The outer boundary of the walkway is approximately how many feet longer than the inner boundary of the walkway?
A.10
B.15
C.20
D.30
E.40

(A) (B) (C) (D) (E)

Question 7 객관식

549. The area of rectangular region R is 50 percent greater than the area of rectangular region S. If the length of R is 25 percent greater than the length of S, then the width of R is what percent greater than the width of S?
A20%
B25%
C75%
D120%
E125

(A) (B) (C) (D) (E)

Question 8 객관식

Color	Number of Cards	Labels
Red	3	*K, S, W*
Blue	3	*S, W, H*
Green	3	*W, H, N*

402. In a set of cards, each card is colored one of three colors and each card is labeled with one letter as indicated in the table. If one card is to be selected at random from the set, what is the probability that the selected card will be blue or labeled with the letter W?
A2/9
B2/9
C4/9
D5/9
E6/9

(A) (B) (C) (D) (E)

Question 9 복수선택

483. If $\sqrt{108} = a\sqrt{b}$, then the sum of a and b could be (a and b are both positive integers)?
Indicate all such numbers.
A9
B15
C29
D45
E109

Indicate all that apply. [A] [B] [C] [D] [E]

Question 10 주관식

673. If x and y are positive integers and x < y, how many different ordered pairs (x, y) are there such that the sum of x and y is 87 and x and y have the same tens digit?

Question 11 객관식

435. In the xy-plane (not shown), a right triangle has its right angle at the origin and has its hypotenuse along the line y=7x-1. If none of the sides of the triangle are vertical, what is the product of the slopes of the three sides of the triangle?
A-7
B-1
C−(1/7)
D1/7
E1

(A) (B) (C) (D) (E)

Question 12 객관식

1319. If n is an integer, what is the least possible value of $3^n+(3)(3^{-n})$
A.1
B.2
C.3
D.3(1/3)
E.4

Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 13 복수선택

968.

Number of colleges visited	0	1	2	3	4 or more
Number of students	3	*x*	24	32	7

The frequency distribution above shows the number of colleges visited by a sample of students. If the median of the numbers of colleges visited is 3, which of the following could be the value of x? Indicate all such values.
A0
B6
C9
D15
E24

Indicate all that apply. [A] [B] [C] [D] [E]

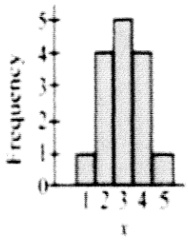
Question 14 객관식

864. Let q be a prime number less than 100. When q is divided by 5, the remainder is 2. When q is divided by 7, the remainder is 6, what is the remainder when q is divided by 8?
A1
B3
C5
D6
E7

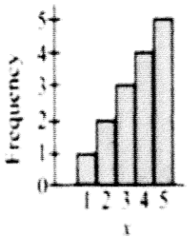
Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

Question 15 객관식

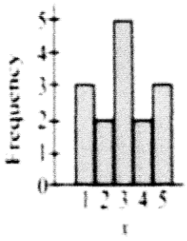
534.



Distribution I



Distribution II



Distribution III

In the three frequency distributions, the variable x has integer values from 1 through 5. For which of the distributions is the average (arithmetic mean) of the 15 values of x equal to the median of the 15 values of x?

A I only
B II only
C I and III only
D II and III only
E I, II and III

(A) (B) (C) (D) (E)